

Md Saifur Rahman

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Hello, I am an enthusiastic engineer deeply passionate about semiconductor devices, nanoelectronics, 2D materials, and applied machine learning, with a focus on advancing ultra-scaled VLSI technologies. I have a fervent interest in academia and have demonstrated outstanding academic performance in my field of study. I want to see myself as an impactful researcher in the future, contributing to technological advancements and making a meaningful impact.

PROFESSIONAL EXPERIENCE

Southeast University

February 2025 - Till Date

Adjunct Lecturer

- Conducted several Labs, theories, exams and OBE based evaluation process
- Courses: Green Electronics, VLSI, Robotics & Automation, Digital Signal Processing, Electronics, Digital Electronics, Power Electronics, Communication Engineering, Electrical service design

ACADEMIC CREDENTIALS

Islamic University of Technology

2024

B.Sc. Electrical and Electronic Engineering

CGPA: 3.84/4.0 (17th out of 117)

Notre Dame College (Dhaka)

2019

Science, HSC (A-level equivalent)

GPA: 5.0/5.0

Al Amin Academy School (Chandpur)

2017

Science, SSC (O-level equivalent)

GPA: 5.0/5.0

RESEARCH INTEREST

Semiconductor devices - 2D materials - VLSI design - nanoelectronics - Machine learning

RESEARCH EXPERIENCE

Comparative TCAD Study of Si and MoS2 Gate-All-Around FETs (Ongoing)

- A TCAD study comparing Si and MoS2 Gate-All-Around FETs (GAAFETs) found that MoS2 channels offer superior electrostatic control, demonstrating higher drive current and better Ion/Ioff ratios across various design parameters.

Evolution of transistor technologies in industry (Ongoing)

- We will examine the evolution of transistor technologies from early devices to advanced MOSFETs, FinFETs, and GAAFETs. It will highlight industrial milestones, scaling limits, and challenges such as short-channel effects, power leakage, and the integration of high-k dielectrics, while discussing future directions in semiconductor innovation.

Residential Load Forecasting and Dynamic Demand Side Management Using Recurrent Neural Networks (THESIS)

- Introduced load forecasting techniques, customer load disaggregation, and load shifting algorithms in a residential area to reduce costs for customers and service providers.

PUBLICATIONS

Journal Articles

- Nadim Ahmed, Mirza Fuad Adnan, Ahmad Shafiullah, Hayder Jahan Parash, **Md. Saifur Rahman**, Irfan Chowdhury Akib, Golam Sarowar, **Digitize-HCD: A Dataset for Digitization of Handwritten Circuit Diagrams**, Data in Brief, 2025, 111315, ISSN 2352-3409, <https://doi.org/10.1016/j.dib.2025.111315>.

Conference Proceedings

- M. S. Rahman**, A. I. Keats, M. A. Kabir, A. Newaz and M. M. Islam, "Survival Analysis of Breast Cancer Patients: A Population-Based Study from SEER," 2023 International Conference on Electrical, Computer and Energy Technologies (ICECET), Cape Town, South Africa, 2023, pp. 1-6, doi: 10.1109/ICECET58911.2023.10389236.

PROJECTS

- Performance Analysis of 4-bit SISO Shift Registers** | *Quartus, Pspice, Microwind* Oct. 2023
- Investigated 4-bit SISO shift registers using circuit analysis, layout design, and timing diagrams, implementing SISO using transmission gate, pass transistor and NAND gates, and comparisons were made.
- Microcontroller-Based DC Motor Speed Control with User Interface** | *AVRDUDESS, Mide* Dec. 2023
- Able to rotate in any direction, maintaining a certain speed that can be controlled by the user. Implemented an emergency stop and limit switch system.
- 8 Bit SAP 1** | *Proteus* 2022
- Developed a 8 Bit SAP-1 on Proteus.
- Survival Analysis of Breast Cancer Patients** | *Machine Learning, Jupyter, Google Colab* Jan. 2023
- Utilizing Kaplan-Meier, Cox regression, and Random Survival Forest models, we estimated breast cancer survival probabilities by exploiting observable relationships within the dataset's attributes.
- Prediction of Phishing Web Page** | *Machine Learning, Jupyter, Google Colab* Aug. 2023
- Achieved 97.95% accuracy and 96.69% precision using ensemble classifier consisting of SVM, Logistic Regression, and Random Forest Classifier algorithm.

TECHNICAL SKILLS

Languages: Matlab, C, Verilog, HTML, LaTeX, Assembly

Data Analysis: Python, Numpy, Pandas, Matplotlib, Seaborn

Design and Simulation: Silvaco Tcad, Atlas, Ltspice

Software: MATLAB, Proteus, Pspice, OrCAD Capture CIS, Tinkercad, Autodesk Eagle, PSIM, MICROWIND, Quartus

TRAININGS

- Bangladesh Power Development Board** | *Industrial trainee* 2023
- Visited Ghorasal power station, Palash, Narsingdi. Get to know details about the operations of a power plant, power generation, and transmission process. Duration: 2 weeks
- Institute of Nuclear Science and Technology** | *Industrial trainee* 2023
- Visited the Nuclear Reactor and all its machinery to gain hands-on experience and knowledge on the operation and functionality of the reactor. Duration: 1 week

AWARDS AND ACHIEVEMENTS

Scholarships

- Received Bangladesh Govt. Scholarship for excellence in HSC 2019
- Received Bangladesh Govt. Scholarship for excellence in SSC 2017
- Received Bangladesh Govt. Scholarship for excellence in JSC 2014

EXTRACURRICULAR ACTIVITIES

Lawn Tennis	2024
• Served as University Lawn Tennis captain • Runner-up in IUT Lawn Tennis Tournament 2024(Double's category) • Runner-up in IUT Lawn Tennis Tournament 2024(Single's category)	
Bangladesh Scouts	2019
• Got President's Scout Award, a prestigious recognition given to scouts who have demonstrated outstanding leadership, service, and dedication to the principles of scouting. It is the highest award attainable within the Bangladesh Scouts organization and is awarded by the honorable President of Bangladesh.	
IUT Mars Rover - Team Avijatrik	2022
• Senior Member - Electrical Team	
PCB Design contest	2021
• Runner-up in the contest of PCB design and fabrication organized by the IEEE Robotics and Automation Society IIUC Student Chapter in coordination with the IEEE IIUC Student Branch.	

REFEREE

Dr. Md. Ashraful Hoque Vice Chancellor Ahsanullah University of Science & Technology (AUST) Former Professor of EEE, IUT Email: mahoque@iut-dhaka.edu Mobile: 01735444474	Md. Ruhul Amin, PhD Professor and Chairman, EEE Southeast University & Former Professor of EEE, IUT Email: ruhul@seu.edu.bd Mobile: 01713228533	Mr. Nadim Ahmed Assistant Professor, Dept. of EEE Islamic University of Technology (IUT) Email: nadimahmed@iut-dhaka.edu Mobile: 01748806097
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